



INDUSTRIAL ATTACHMENT REPORT

BY

MARY WAINAINA

REG. NO: 21/06025

BACHELOR OF SCIENCE IN SOFTWARE DEVELOPMENT (BSD)

KCA UNIVERSITY

ATTACHMENT ORGANIZATION: AGILE BUSINESS SOLUTIONS LTD

INDUSTRY SUPERVISOR: ROY KIPCHUMBA

PERIOD: SEPTEMBER – NOVEMBER 2025

ABSTRACT

This report presents a summary of my industrial attachment experience at Agile Business Solutions Ltd, where I was attached from September to November 2025 as a backend developer trainee. The attachment was part of the requirements for the Bachelor of Science in Software Development program at KCA University.

During the three-month period, I participated in practical software development activities focused on .NET backend development. I worked on two main projects , a School Management System and an Intern/Attaché Management System. The internship enabled me to apply theoretical knowledge gained in class to real-world problems, develop teamwork and communication skills, and gain hands-on experience in system design, API development, and integration with frontend systems developed in React.

Through this experience, I improved my understanding of the software development lifecycle (SDLC), particularly in Agile methodologies, version control using Git, RESTful API design, and database management using Microsoft SQL Server.

DECLARATION

I hereby declare that the information contained in this document is my original work based on truth and accurately reflects the activities and duties that I undertook during my industrial attachment. The information has been presented in accordance with academic integrity and ethical conduct.

I further declare that, as required by these rules and conduct, I have personally written this report. Therefore, I affirm that this material is my own original work.

STUDENT: MARY WAINAINA

REG. NO: 21/06025

SIGNATURE:

DATE:

INDUSTRY SUPERVISOR: ROY KIPCHUMBA

SIGNATURE:

DATE:

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Agile Business Solutions Ltd for giving me the opportunity to complete my industrial attachment in such a supportive and professional environment.

Special appreciation goes to my industry supervisor, Roy Kipchumba, for her guidance, constructive feedback, and encouragement throughout the attachment period.

I also thank my university supervisor, Purity Mureithi and the School of Technology, KCA University, for facilitating this attachment program, which provided me with essential practical exposure to software development.

Finally, I thank my team members and colleagues at Agile Business Solutions Ltd for their collaboration and assistance during various stages of development.

Contents

ABSTRACT.....	2
ACKNOWLEDGEMENT	4
1. CHAPTER ONE	6
1.1. INTRODUCTION	6
1.2. OBJECTIVES OF THE INDUSTRIAL ATTACHMENT.....	6
1.3. BACKGROUND OF AGILE BUSINESS SOLUTIONS LTD	7
1.4. ORGANIZATIONAL STRUCTURE	7
2. CHAPTER TWO	8
2.1. HOST DEPARTMENT – SOFTWARE DEVELOPMENT	8
2.2. KEY FUNCTIONS OF THE DEPARTMENT	8
2.3. STAFF ESTABLISHMENT	8
2.4. DUTIES AND RESPONSIBILITIES	9
3. CHAPTER THREE	10
3.1. MAJOR ACTIVITIES UNDERTAKEN	10
3.2. SKILLS GAINED	11
3.3. CHALLENGES FACED	11
4. CHAPTER FOUR	13
4.1. RECOMMENDATIONS.....	13
4.2. SUGGESTIONS TO THE UNIVERSITY.....	13
4.3. CONCLUSION	13

1. CHAPTER ONE

1.1. INTRODUCTION

The industrial attachment is a crucial component of the Bachelor of Science in Software Development program. It provides students with an opportunity to gain hands-on experience, understand the work environment, and bridge the gap between classroom learning and industry practice.

My attachment took place at **Agile Business Solutions Ltd**, a technology company that focuses on software design, development, and business automation solutions. I was placed in the **Software Development Department**, where I participated in the design and backend development of web-based management systems using **ASP.NET Core**, **Entity Framework**, and **SQL Server**.

Through this experience, I enhanced my technical, analytical, and teamwork skills while applying classroom concepts in real-world software projects.

1.2. OBJECTIVES OF THE INDUSTRIAL ATTACHMENT

The main objectives of the attachment were:

- To gain hands-on experience in .NET backend software development.
- To apply classroom knowledge in real-world project environments.
- To understand the structure and workflow of a software development team.
- To improve skills in database design, API development, and project collaboration using Git and Agile methodologies.
- To develop soft skills such as teamwork, time management, and communication in a professional setup.

1.3. BACKGROUND OF AGILE BUSINESS SOLUTIONS LTD

Agile Business Solutions Ltd is a Kenyan-based software company that provides digital and business process automation solutions to organizations across different sectors. The company specializes in custom software development, enterprise systems, web applications, and business intelligence tools.

The company's core philosophy is built around Agile methodology, focusing on continuous improvement, iterative development, and collaboration between clients and development teams. Agile Business Solutions has a diverse team of developers, project managers, and designers who work together to deliver scalable and user-friendly systems.

The company's main products include Management Systems, Human Resource and Payroll Systems, Finance management systems and other enterprise applications tailored for institutions and SMEs.

1.4. ORGANIZATIONAL STRUCTURE

The company operates under a flat, collaborative structure that encourages teamwork and open communication. The typical structure includes:

- ✓ **Chief Executive Officer (CEO)** – Oversees overall operations and business direction.
- ✓ **Project Manager / Team Lead** – Coordinates projects, monitors progress, and ensures deliverables meet client expectations.
- ✓ **Software Developers (Backend and Frontend)** – Design, develop, and maintain system components.
- ✓ **UI/UX Designers** – Handle interface design and user experience.
- ✓ **Quality Assurance (QA) Engineers** – Conduct system testing and bug tracking.
- ✓ **Interns / Trainees** – Assist in development tasks while learning under supervision.

2. CHAPTER TWO

2.1. HOST DEPARTMENT – SOFTWARE DEVELOPMENT

I was attached to the Bespoke Department, which is responsible for the design and implementation of software products for clients. The department focuses on web-based solutions using technologies such as .NET Core, React.js, SQL Server, and RESTful APIs.

During my attachment, the department handled several projects, I was assigned School Management System and the Intern/Attaché Management System. My work was mainly focused on the backend development aspect of these systems.

2.2. KEY FUNCTIONS OF THE DEPARTMENT

- Designing and developing software systems for clients.
- Maintaining and upgrading existing systems.
- Database design and optimization.
- Collaborating with the frontend team to integrate APIs.
- Testing and debugging applications before deployment.
- Using Agile methodologies for sprint planning, daily stand-ups, and progress reviews.

2.3. STAFF ESTABLISHMENT

The department consisted of:

- Project Manager / Team Lead
- Backend Developers
- Frontend Developers

- QA Engineer
- 5 Attacheés (including myself)

The backend team worked with .NET Core and Entity Framework, while the frontend team used React.js and Tailwind CSS.

2.4. DUTIES AND RESPONSIBILITIES

My main responsibilities included:

- ✓ Developing and maintaining backend APIs using ASP.NET Core.
- ✓ Implementing database models and relationships in SQL Server.
- ✓ Participating in code reviews and debugging sessions.
- ✓ Collaborating with frontend developers to integrate endpoints.
- ✓ Writing documentation for API routes and project modules.
- ✓ Testing modules using Postman to ensure data integrity and endpoint reliability.
- ✓ Using GitHub for version control and team collaboration.

3. CHAPTER THREE

3.1. MAJOR ACTIVITIES UNDERTAKEN

I worked on two main projects:

1. School Management System

This system aimed to automate school operations such as student registration, class management, and grading. My role focused on designing and implementing the backend logic for student records, subjects, and grading modules using ASP.NET Core MVC. I developed RESTful APIs that the frontend team could consume for managing school data.

2. Intern/Attaché Management System

This was a group project involving five members, three backend developers and two frontend developers. The project aimed to streamline the management of interns within an organization, including registration, assignments, attendance, and evaluation.

- I specifically handled the Learning & Assignment Management Module, which managed intern tasks, submissions, and supervisor feedback.
- I created endpoints for assignment upload, feedback retrieval, and progress tracking.
- Other team members worked on User Management and Absentee Management Modules.
- The frontend team developed the user interface in React.js, which was integrated with our backend APIs.

During collaboration, we held daily stand-up meetings, used GitHub for version control, and followed the Agile development approach through sprints and reviews.

3.2. SKILLS GAINED

- ✓ Proficiency in .NET Core and C# programming.
- ✓ Practical experience in designing and developing RESTful APIs.
- ✓ Understanding of Agile methodologies and sprint-based teamwork.
- ✓ Experience with database design, relationships, and stored procedures in SQL Server.
- ✓ Improved knowledge in API testing using Postman.
- ✓ Enhanced version control and collaboration using GitHub.
- ✓ Communication and problem-solving skills in a professional software team setting.

3.3. CHALLENGES FACED

- Integration issues during API consumption by the React frontend team.
- Version control conflicts during team collaboration on GitHub.
- Limited time for completing project milestones due to sprint deadlines.
- Occasional database synchronization errors between development environments.

However, these challenges strengthened my debugging, time management, and collaboration skills.

Strong Points:

- ✓ The company offered a very collaborative and learning-friendly environment where team members were open to guidance and sharing knowledge.
- ✓ I gained real experience with Agile methodology, daily stand-ups, and sprint reviews, which helped me understand how modern software teams operate.

- ✓ I had access to professional development tools such as GitHub, Visual Studio, SQL Server, and Postman, which enhanced my technical skills.
- ✓ The mentorship from my supervisor and colleagues strengthened my coding discipline, version control practices, and backend logic design.

4. CHAPTER FOUR

4.1. RECOMMENDATIONS

- ✓ Maintain hybrid mentorship : allowing interns to explore both backend and frontend components for better full-stack understanding.

4.2. SUGGESTIONS TO THE UNIVERSITY

- Strengthen partnerships between KCA University and tech companies for consistent placements.
- Introduce mid-attachment evaluations to track students' progress.
- Provide a standard template and guidance for documentation and reporting.

4.3. CONCLUSION

The industrial attachment at Agile Business Solutions Ltd was an eye-opening and enriching experience. It gave me an opportunity to transform classroom theory into real-life software development practice.

I acquired hands-on skills in .NET backend development, teamwork, and Agile-based collaboration. The experience boosted my confidence and strengthened my understanding of how software projects are planned, built, and deployed in professional environments.

I am grateful to Agile Business Solutions Ltd, my supervisors, and KCA University for facilitating this valuable experience that has greatly prepared me for my future career in software development.